

INTERNSHIP REPORT ON

INTEPS26IPM003

Yeshwantrao Chavan College of Engineering

**Bachelor of Technology
In
Computer Science & Engineering**



Session 2025-26

Submitted by:

Pranav Satish Nilawar

Supervised by:

Faculty Supervisor : Prof. Snehal Dongre

Industry supervisor : Ms. Yalin Maria

Nagar YuwakShikshanSanstha's

YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING,
(An Autonomous Institution Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur)

NAGPUR – 441 110

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CERTIFICATE OF APPROVAL

Certified that the Internship project entitled” INTEPS26IPM003 “has been successfully completed by Pranav Satish Nilawar during session 2025-26

Faculty Supervisor

Signature : _____

Name : Prof. Snehal Dongre

Designation: Assistance Professor - YCCE

Industry Supervisor

Signature : _____

Name : Ms. Yalin Maria

Designation: CPO L&D GenC Training – Cognizant

Technology Solutions Corporation

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I, Pranav Satish Nilawar would like to convey my gratitude to Yeshwantrao Chavan College of Enginneing *Yalin Maria ,CPO L&D GenC Training, Cognizant Technology Solutions Corporation* for emphasizing on the Semester Internship Program and giving me the platform to interact with industry professionals. I would also like to thank *Prof. Snehal Dongre and Dr. Lalit Damahe* for giving me the opportunity to work on the prestigious Internship.

I extend my warm gratitude and regards to everyone who helped me during my internship

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Introduction of the Report

The internship program is an essential part of the Bachelor of Technology curriculum, designed to provide students with practical exposure to real-world industry environments. This report presents a detailed account of the learning, experiences, and contributions made during the internship under the program **INTEPS26IPM003**.

The primary objective of this internship was to bridge the gap between theoretical knowledge gained during academic studies and its practical application in the industry. Through this opportunity, I was able to understand how professional organizations function, how projects are executed in real time, and how teamwork and communication play a vital role in successful project delivery.

During this internship, I worked in a professional development environment where I was exposed to modern tools, technologies, and workflows used in the software industry. This experience allowed me to enhance my technical skills, improve my problem-solving abilities, and gain a deeper understanding of software development practices. I also learned the importance of writing clean, efficient, and maintainable code while adhering to industry standards.

The internship not only focused on technical development but also contributed significantly to my personal growth. I developed better communication skills, learned how to collaborate effectively within a team, and adapted to a structured work environment. Time management, responsibility, and continuous learning were key aspects that I practiced throughout the internship period.

This report outlines the overall journey of my internship, including an overview of the organization, details about the project work, technologies used, challenges faced, and the learning outcomes achieved. It also highlights how this experience has contributed to my professional development and prepared me for future career opportunities.

The transition from an academic environment to a professional industrial setting is a crucial phase for any engineering student. This internship report documents a comprehensive professional training journey undertaken at Cognizant Technology Solutions within the **INTEPS26IPM003** program. As a student of Computer Science & Engineering at Yeshwantrao Chavan College of Engineering (YCCE), this experience served as a strong bridge between theoretical concepts learned in the classroom and the practical demands of the IT industry.

1.1 Purpose and Objectives

The primary objective of this internship was to gain hands-on experience in IBM webMethods (Software AG Designer) and understand industry-level software development practices. In today's era of rapid digital transformation, acquiring such skills is essential for building scalable and efficient applications.

The internship was designed with the following key goals:

- **Technical Proficiency:** To understand the complete development lifecycle, including flow services, messaging, adapter notification.
- **Industry Practices:** To gain exposure to professional workflows such as Agile methodologies, version control systems, and structured development approaches.
- **Holistic Development:** To improve both technical and interpersonal skills required for working effectively in a corporate environment.

1.2 Training Methodology

A unique aspect of this internship was the focus on self-learning combined with practical implementation. The training encouraged an active learning approach where concepts were understood through hands-on practice, assignments, and continuous evaluation.

This approach helped in developing problem-solving skills, independent thinking, and the ability to adapt to new technologies efficiently.

1.3 Scope of the Technical Journey

The learning process during the internship was structured in a progressive manner, starting from foundational concepts and moving towards advanced development practices. It included:

1. Building a strong foundation in introduction to webMethods
2. Understanding development concepts
3. Learning adapters, messaging, web services, rest services
4. Gaining exposure to deployment concepts and real-world application workflows

This structured approach ensured a complete understanding of IBM webMethods.

1.4 Practical Experience

An important part of the internship was working on a application, which helped in applying the learned concepts in a real-world scenario. The project involved designing, developing, and integrating different components of an application.

This experience helped me understand:

- Requirement analysis and implementation
- Application structure and workflow
- Debugging and optimization techniques
- Importance of teamwork and coordination

This report provides a comprehensive overview of the internship journey, including the skills acquired, challenges faced, and knowledge gained throughout the program. It reflects not only the technical growth achieved during the internship but also the development of professional skills required in the industry.

Overall, the internship experience has been highly valuable in preparing me for future opportunities and strengthening my foundation as a software professional.

Overview of the Internship providing Organization

2.1 Introduction to the Organization

The internship was carried out at Cognizant Technology Solutions, a globally recognized multinational corporation that provides IT services, consulting, and business process outsourcing solutions. Headquartered in Teaneck, New Jersey, USA, Cognizant has established a strong global presence, serving clients across industries such as healthcare, banking, retail, manufacturing, and technology.

Cognizant is widely known for its strong focus on digital transformation. It helps organizations modernize their technology infrastructure, optimize business processes, and adopt innovative digital solutions. The company offers a wide range of services including application development, cloud computing, data analytics, artificial intelligence, and enterprise-level solutions. Its consistent commitment to quality, innovation, and customer satisfaction has made it one of the leading IT service providers worldwide.

2.2 Corporate Profile: Engineering Modern Businesses

Cognizant Technology Solutions is a global leader in professional services, dedicated to helping organizations build stronger, more agile, and future-ready businesses. Established in 1994, the company has grown from a small technology unit into a Fortune 500 organization and a major contributor to the global IT ecosystem.

The company combines deep industry knowledge with advanced technological capabilities to deliver impactful solutions. It focuses on enabling clients to modernize systems, improve operational efficiency, and enhance user experiences. This ability to integrate technology with business strategy makes Cognizant a trusted partner for enterprises across the world.

2.3 Global Presence and Indian Ecosystem

Although Cognizant is headquartered in the United States, a significant part of its operations is based in India. The company has major delivery centers in cities such as Chennai, Pune, Bangalore, and Nagpur. These centers play a crucial role in large-scale software development, testing, and digital transformation initiatives.

Being part of this ecosystem provided exposure to a highly professional and collaborative environment. With a workforce of thousands of skilled professionals globally, Cognizant offers a diverse and inclusive work culture that enables knowledge sharing, innovation, and continuous learning.

2.4 Internship Program: GenC Framework

The internship was conducted under Cognizant's GenC (Generation Cognizant) program,

which is specifically designed to develop industry-ready talent. This program focuses on building strong technical foundations along with essential professional skills.

The GenC framework is structured into different levels such as GenC, GenC Pro, and GenC Next, each catering to different skill levels and career paths. It emphasizes practical learning, problem-solving, and real-world application of knowledge rather than purely theoretical understanding.

2.5 Training Methodology: Learner-Centric Approach

A key highlight of the internship was its modern learning approach based on Learner Autonomy. This method encourages interns to take ownership of their learning process.

The training followed a Flipped Classroom Model, which included:

- Pre-learning of concepts through online platforms
- Hands-on coding sessions and practical assignments
- Regular assessments such as Integrated Capability Tests (ICTs)
- Continuous feedback and improvement

This approach ensured that learning was active, practical, and aligned with industry requirements.

2.6 Exposure to Modern Technologies

During the internship, there was significant exposure to modern tools, technologies, and development practices. The program focused on real-world problem-solving and the use of current industry technologies.

Additionally, Cognizant is actively investing in emerging technologies such as Generative AI (GenAI). The integration of AI-based tools into the learning process helps developers enhance productivity, improve code quality, and build innovative solutions. This forward-thinking approach ensures that interns are prepared for future technological advancements.

2.7 Work Culture and Professional Environment

Cognizant follows a highly professional and collaborative work culture. Teamwork, innovation, and continuous improvement are strongly encouraged within the organization.

Interns are provided with:

- Guidance from experienced mentors
- Access to structured learning resources
- Opportunities to work on practical assignments and projects

This supportive environment helps in building both technical competence and confidence in handling real-world challenges.

2.8 Industry Practices and Processes

The organization emphasizes adherence to industry-standard practices, which are essential for efficient software development. These include:

- Agile methodologies for project management
- Version control systems for code management
- Structured development workflows
- Testing and debugging practices for quality assurance

Understanding these processes provided valuable insight into how professional software systems are developed and maintained.

2.9 Mission and Core Values

Cognizant operates with a clear mission: to leverage technology and innovation to strengthen client businesses. This mission is supported by a strong set of core values that guide every employee:

- Encouraging innovative and independent thinking
- Making data-driven decisions
- Continuously striving for improvement
- Promoting teamwork and collaboration
- Maintaining integrity and professionalism

These values play a vital role in shaping the organization's culture and success.

2.10 Overall Experience

Overall, the internship at Cognizant provided valuable exposure to the corporate environment, industry practices, and modern technologies. It significantly enhanced my technical knowledge and professional skills.

This experience helped me understand real-world software development, improve my problem-solving abilities, and prepare myself for future career opportunities in the field of Computer Science and Engineering.

About the Internship

3.1 Introduction

As part of the internship program INTEPS26IPM003 at Cognizant Technology Solutions, I worked on the IBM webMethods Integration Platform — a robust, enterprise-grade middleware and integration solution developed by IBM (originally by Software AG). The internship was designed to provide hands-on exposure to integration technologies, Business Process Management (BPM), messaging systems, adapters, and modern cloud-based integration using webMethods.io.

IBM webMethods is widely used across industries to connect disparate applications, automate business workflows, enable real-time data exchange, and support B2B trading partner communications. The platform includes a rich suite of tools: Integration Server (IS), My webMethods Server (MWS), Broker, Universal Messaging (UM), Trading Networks (TN), and the cloud-native webMethods.io.

The training program was structured into multiple modules progressing from foundational architecture concepts to advanced topics such as JAVA services, Web Services, REST APIs, Adapters, and B2B integration, providing a complete view of enterprise integration using the webMethods platform.

3.2 Overview

IBM webMethods is a comprehensive integration platform that enables organisations to connect applications, automate processes, and facilitate B2B communications. It is one of the most mature and feature-rich middleware platforms available in the enterprise market.

3.2.1 Design Time vs. Run Time vs. Admin

webMethods distinguishes between three operational phases: Design Time (where services and flows are created using Designer), Run Time (where Integration Server executes services, processes messages, and handles transactions), and Administration (where MWS and IS Admin pages are used to monitor, configure, and manage the platform).

3.2.2 Core Platform Components

Component	Description
Integration Server (IS)	The core runtime engine that hosts packages, services, and connectors. It processes inbound/outbound requests and orchestrates integrations.
My webMethods Server (MWS)	A web-based administration and monitoring portal for managing services, processes, and system health.
Broker / Universal Messaging (UM)	Provides publish-subscribe and point-to-point messaging capabilities for reliable, asynchronous communication.
webMethods Designer	An Eclipse-based IDE used to design, develop, and deploy integration services, flow services, and Java services.
Trading Networks (TN)	Manages B2B partner profiles, document types, and processing rules for EDI, XML, and flat-file exchanges.
webMethods.io	A cloud-native integration platform supporting workflow automation, API management, and hybrid integrations.
JDBC Adapter	Enables connectivity to relational databases through standardized adapter services and notifications.

3.3 Objectives

The primary objectives of this internship were:

- To understand the architecture and components of IBM webMethods Integration Platform, including IS, MWS, Broker, and UM.
- To gain practical experience in developing Flow Services using steps such as MAP, BRANCH, INVOKE, LOOP, SEQUENCE, REPEAT, and EXIT.
- To learn pipeline management, data types, and the usage of built-in WmPublic services.
- To understand messaging paradigms including pub-sub, request-reply, JMS Topics/Queues, and Triggers.
- To work with file handling mechanisms such as FlatFiles, FTP, and File Polling.
- To develop and consume Web Services (SOAP/WSDL) and RESTful APIs within the webMethods environment.
- To configure and use JDBC Adapters for database connectivity and notifications.
- To understand B2B integration concepts using Trading Networks (TN), including partner profiles, document types, and processing rules.

- To explore cloud-based integration using webMethods.io, including workflow automation and messaging.
- To complete hands-on assignments and a technical evaluation to demonstrate competency.

3.4 Scope of the Internship

The scope of this internship spanned the full lifecycle of enterprise integration development using IBM webMethods. The training was comprehensive and covered the following broad areas:

3.4.1 Platform Architecture & Administration

Understanding the physical and logical layout of webMethods installations, server configuration, startup/shutdown procedures, server and error log management, naming conventions, and IS admin page operations.

3.4.2 Core Development (Flow Services)

Development of integration logic using Flow Services with detailed exploration of all flow steps, pipeline debugging, service caching, error handling (try/catch, ISRuntimeException retry), and use of WmPublic services.

3.4.3 Messaging & Eventing

Hands-on implementation of Broker and Universal Messaging concepts, JMS integration, trigger configuration (local, broker, JMS), serial and concurrent processing, and canonical document design.

3.4.4 File Handling

Working with FlatFiles (fixed-width and delimiter-based), FTP connectivity, and File Polling for event-driven file-based integrations.

3.4.5 Adapter Integration

Configuring JDBC adapters, working with connection pools, performing Select/Insert/Dynamic/Custom/Batch adapter services, managing transactions, and setting up JDBC notifications.

3.4.6 Web Services & REST APIs

Creating and consuming SOAP-based Web Services (understanding WSDL, operations, bindings, ports, handlers, and fault handling), developing Java Services, and building REST service endpoints.

3.4.7 B2B Integration

Configuring Trading Networks (TN) with enterprise/partner profiles, document types, processing rules, and monitoring TN transactions for XML and flat-file exchanges.

3.4.8 Cloud Integration — webMethods.io

Registering and using webMethods.io to create cloud workflows, REST/SOAP API-based flows, pub-sub messaging with Topics/Queues, trigger-based workflows, and error handling flows.

3.5 Training Curriculum & Topics Covered

3.5.1 Introduction to webMethods Architecture

- webMethods overview: Integration and BPM concepts
- Design Time, Run Time, and Admin components
- IS, MWS, Broker, Adapter, Designer, Packages, and IS elements
- How to use Designer and create Packages
- Physical location of installations, packages, and configurations
- Naming conventions and standards
- Server/Error log inspection in IS
- IS directives: DEFAULT, INVOKE, WS, WEB, REST, SOAP
- Start/Stop IS, Broker, UM, and MWS (Windows Service/App and Unix process)
- Admin page details for IS and MWS

3.5.2 Development Concepts

- Flow Services and Steps: MAP, BRANCH, INVOKE, TRANSFORMERS, SEQUENCE, LOOP, REPEAT, EXIT
- Data types and pipeline concepts; save/restore pipeline services
- Service cache, debugLog service, clearPipeline
- appendToDocumentList and appendToStringList
- Pipeline debug property and retry on ISRuntimeException
- Try/catch blocks in flow services
- Flow step properties and flow service properties exploration
- Direct Mapping, Conditional Mapping, Substitutions (Variable and Global Variable)
- Debugging flow services; dropping variables
- WmPublic package services exploration
- Wrapper Services; User/Group/ACL setup
- Package dependency management and package install/release across environments
- MWS Monitoring for flow service and pipeline data
- contextId, parentContextId, rootContextId; service threads
- Invoking a service from browser or external client
- Startup, shutdown, and replicate services
- Copy/paste IS elements; reload/refresh a package
- XSLT service (XML to HTML); Regular Expressions; webMethods Query Language (WQL)
- Applying content-type for string variables

3.5.3 File Handling

- FlatFile processing (fixed and delimited formats)
- FTP connectivity and operations
- File Polling for event-driven integrations

3.5.4 Messaging (Native & JMS)

- IS and Broker communication: Broker Monitor, Broker Server, Broker, MetaData, Runtime Elements
- Broker Territory and Broker Gateways
- pub-sub and request-reply messaging models
- JMS Provider (Broker/UM), JMS Topics and Queues
- Sending and receiving JMS messages using IS built-in elements
- Canonical Document design
- Local/Broker Trigger and JMS Trigger configuration
- Subscribing to multiple documents in a Trigger; invoking multiple services from a Trigger
- Serial vs. concurrent processing with Triggers
- Deleting, disabling, suspending retrieval, and suspending processing of a Trigger

3.5.5 Adapters

- Introduction to Adapters; JDBC Adapter, Drivers, and Connections
- Package management for adapter connections, services, and notifications
- Select, Insert, Dynamic, Custom, and Batch adapter service templates
- Transaction management: boundaries, types, start/commit/rollback services
- Calling/invoking adapter services from a flow service
- Wrapper service for adapter service with cache enabled
- Connection pool parameters: min/max pool, block/expire/transaction timeout, connection alias

- JDBC Basic Notification and JDBC Insert Notification

3.5.6 JAVA / Web Services / REST Services

- Java Services development and debugging/tracing
- WebServices: WSDL contents, SOAP, WSDD, Provider/Consumer, Operations, Binders, Ports, Endpoints
- SOAP fault handling and handlers
- REST Services introduction

3.5.7 B2B Basics (Trading Networks)

- TN concepts: TN, B2B, Enterprise, Organisation, Company, Partners
- Creating a Profile, Document Type, and Processing Rule
- Sending/receiving XML and Flat files via TN
- Monitoring and resubmitting/reprocessing TN transactions
- eStandards supported in webMethods B2B

3.5.8 webMethods.io Basics

- Registering and accessing wM.io (free or Cognizant tenant)
- Creating basic workflows with logger, loop, switch, and transform utilities
- Workflow as REST API, consuming REST APIs, and providing flow service as SOAP API
- Publish/subscribe using Topics/Queues and Workflows (Internal Messaging Provider)
- Triggers: webhook, polling, HTTPRequest; project parameters, debug, and monitor
- Error handling in workflow and flow service integrations
- Centrasite and API Gateway Basics Theory

3.6 Technology Stack

The following technologies, tools, and platforms were used during the internship:

Category	Technology / Tool	Purpose
Integration Platform	IBM webMethods IS	Core runtime for hosting and executing services
IDE / Design Tool	webMethods Designer	Eclipse-based IDE for service development
Admin & Monitoring	My webMethods Server (MWS)	Web-based monitoring and administration
Messaging	Broker / Universal Messaging	Pub-sub and JMS messaging
Database Connectivity	JDBC Adapter	Relational DB integration (Oracle/MySQL/SQL Server)
Web Services	SOAP / WSDL / REST	Service-oriented and REST-based integrations
Programming Language	Java	Java Services within IS
Data Formats	XML, JSON, Flat Files, EDI	Multi-format data exchange
B2B Platform	Trading Networks (TN)	Partner onboarding and B2B document exchange
Cloud Integration	webMethods.io	Cloud-native workflow and API integration
API Management	API Gateway / Centrasite	API publishing, governance, and management
OS Platforms	Windows / Unix/Linux	Deployment and process management environments

3.7 Learning Outcomes

Upon completion of this internship training, the following competencies were acquired:

3.7.1 Technical Skills

- Ability to design, develop, and deploy integration services on IBM webMethods Integration Server.
- Proficiency in building flow services using all major flow steps and handling pipeline data.
- Hands-on experience with asynchronous messaging using Broker, Universal Messaging, and JMS.
- Skill in configuring JDBC Adapters for database connectivity, CRUD operations, and notifications.
- Capability to develop and consume SOAP-based Web Services and RESTful APIs.
- Understanding of B2B integration using Trading Networks for XML and flat-file document exchange.
- Experience with cloud-native integration using webMethods.io for workflow automation.

3.7.2 Platform & Tool Proficiency

- Navigating and configuring IS Admin pages, MWS monitoring dashboards, and Designer IDE.
- Managing packages, services, ACLs, user/group permissions, and server configurations.
- Using IS built-in utilities: debugLog, clearPipeline, appendToDocumentList, pipeline save/restore.
- Monitoring and debugging services using MWS, pipeline debug property, and IS logs.

Conclusion

The internship carried out under the program INTEPS26IPM003 at Cognizant Technology Solutions has been a highly valuable and enriching experience, both technically and professionally. This internship provided me with the opportunity to bridge the gap between academic knowledge and real-world industry practices, allowing me to understand how software systems are designed, developed, and maintained in a professional environment.

Throughout this internship, I gained hands-on experience in working with modern technologies and development tools while contributing to the internship in IBM webmethods. This experience enabled me to understand the complete software development lifecycle, starting from development concepts, messaging, and adapter notification. It also gave me practical exposure to concepts such as modular architecture.

One of the most significant aspects of this internship was the experience of working in a collaborative team environment. Coordinating with team members, following structured workflows, and managing tasks efficiently helped me develop essential professional skills such as communication, time management, and problem-solving. The use of version control systems and Agile methodologies further enhanced my understanding of how large-scale projects are executed in the industry.

In addition to technical growth, this internship contributed to my personal development. I learned how to adapt to new challenges, approach problems logically, and continuously improve my skills through self-learning and practice. The challenges faced during the development process helped me become more resilient and confident in handling complex tasks.

Overall, this internship has played a crucial role in shaping my understanding of the IT industry and preparing me for future career opportunities. The knowledge and experience gained during this period will serve as a strong foundation for my professional journey in the field of Computer Science and Engineering.